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Objective → Circuit that can find 1s complement of a 4 bit number and modify the circuit to find 2s complement of the no.

Requirement → Logisim Simulator.

Theory.

1's complement

The 1's complement of a binary number is obtained by inverting all bits of the number.

That is:

- 0 is replaced by 1
- 1 is replaced by 0

This can be implemented using NOT Gates.

2's complement.

The 2's complement of a binary number is obtained by:

1. Taking the 1's complement, and
2. Adding 1 to the result.

This is commonly used to represent negative number in binary system

Procedure.

Part A: 1's complement.

1. Open logisim and create a new circuit
2. Place a 4-bit input pin
3. Use a splitter to divide the input into 4 individual bits.
4. Connect each bit to a NOT Gate.
5. Combine outputs and connect to output pins.
6. Verify that each bit is inverted.

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Part B: 2's complement.

1. Use the output of the 1's complement circuit.
2. Connect it to a 4 bit adder.
3. Set the second input of the adder as 0001 (constant = 1).
4. Observe the output of the adder.
5. Verify that result is the 2's complement of the input.

Result.

The circuit was successfully designed and implemented in logisim

- The 1's complement was obtained by inverting all bits using Not gates.
- The 2's complement was obtained by adding 1 to the 1's complement using a a 4 bit adder.

Hence, The experiment was completed successfully.

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Objective -> to design a 4 ~~with~~ bit arithmetic circuit.

Requirement: Logisim Simulator

Theory: A 4 bit arithmetic circuit is a combinational circuit that perform arithmetic operations on binary no:

it mainly uses:
: Full Adders
: multiplexers (mux)
: Control (Selection) lines

Working principle:
The circuit performs operations based on Selection inputs (S_1 S_0) and carry input (C_{in})

Each bit operation is performed using a 4 Full adder, where:

- One input is the actual data (A)
- The second input is selected using mux
- Third input is carry (C_{in})

Procedure.
: Open logisim and create new circuit.
: place two - 4 bits input (A and B)
: Use Splitters to separate bits.
: Connect each bit to a 4: 1 MUX

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5. Provide inputs ~~to~~ to MUX
   - B
   - B' (using NOT Gates)
   - 0
   - 1
6. Use Selection lines (S_1, S_0) to control MUX output.
7. Connect MUX outputs to Full adders.
8. Connect A inputs directly to Full adders.
9. Provide carry input (C_{in}) to the First adder.
10. Connect Outputs of a Full adders to output pins.
11. Verify different operations by changing Selection lines.

Result:

The 4-bit airthemetic circuit was successfully designed and implemented using Logisim The circuit performed multiple arithmetic operation such as:

- Addition
- Substraction
- incremenet.
- Transfer

based on the Selection inputs.

Hence the experiment was completed successfully.

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